

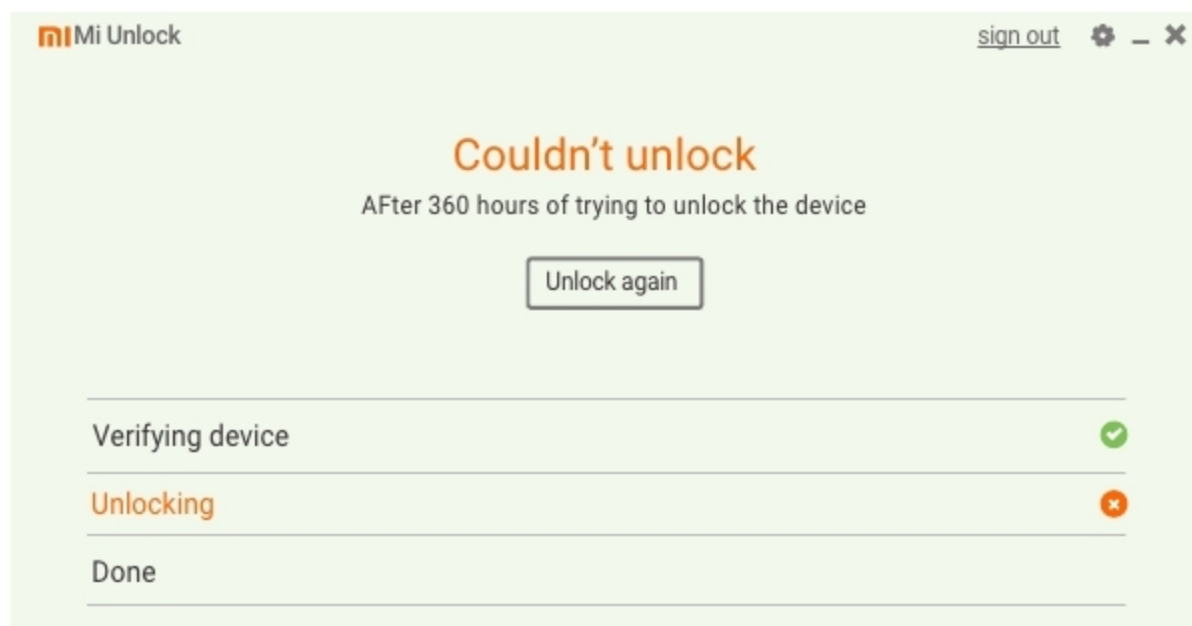


Figure 2: Could not unlock

process to boot the operating system installed in the file system.

Most bootloaders are locked by vendors to make sure the user sticks to the operating system specifically designed by the vendor for that particular device. With a locked bootloader, it is impossible to flash a custom ROM and a wrong attempt may brick the device. Again, this is one of the security features provided by the vendor.

The bootloader can be unlocked in two ways — one is the official method provided by the vendor, and the other is the unofficial method which may lead to a bricked device.

Installing Pixel Experience on a Xiaomi device

The previous section covered concepts that we needed to be clear about before flashing a custom ROM. Now let's go through a step by step procedure to install a custom ROM on any Android device. We're working specifically on a Redmi device and this is based on my own experience.

Here are some points to remember before unlocking the bootloader:

- Take the phone's backup on a PC/laptop (you are unlikely to lose any data, and this step is just a precaution).
- Unlocking the bootloader voids the warranty.
- Make sure that the zip file of the Android ROM is downloaded to the device's internal memory or SD card.

- Make sure that the bootloader of the device is unlocked after the unlock process completes, because a wrong attempt may brick the device.

Steps to follow on a laptop/PC

1. On your laptop/PC, navigate to <https://en.miui.com/unlock/> and click on the *Unlock Now* button.
2. Log in to the MI account with the credentials you used to log into your device.
3. Remember the credentials, since this is the most important step.
4. As per the new MI unlock bootloader method, you don't need permission from MI. To download the MIUI *Unlock* application, simply click on the button. The size is around 55MB.
5. Go to where we downloaded the MIUI *Unlock* application in Step 4, and double click on *miflash_unlock.exe*.
6. Log in using the MI account, which is used in Step 2.
7. Make sure that the device is properly connected to the PC using a USB cable (the status will be shown in the application).

Steps to follow on a mobile phone

1. Go to *Settings->About phone*.
2. Tap five times on the MIUI version; it will enable the *Developer option* on your device.
3. Go to *Settings->Additional Settings->Developer options* and tap on *OEM unlocking*. Do not enable this but click on the text to go inside.

4. Enter your password/PIN (whichever is set to the device for confirmation) and enable the option.
5. Go to *Settings->Additional Settings->Developer options*, and then go to *MIUI status* and tap on *Add account*. This step will add the MI account to unlock the bootloader. Sometimes the account does not get added, in which case, restart the phone.
6. Enable USB debugging from *Settings->Additional Settings->Developer options*.
7. Switch off the phone and press the *Power* button and the 'volume down' key simultaneously.
8. Connect the USB cable to the device and laptop/PC.

After completing these steps on the mobile, repeat the following steps on your PC/laptop

1. Click on the *Unlock* button.
2. A dialogue box with the following message will appear: "Unlocking the phone will erase all phone data. Do you still want to continue to unlock the phone?" Click on *Unlock anyway*.
3. You will see the message shown in Figure 2, if it did not unlock. There is a time specified after which you may try again. The time varies between 24 hours to 360 hours. In my case it was 360 hours, which is nothing but 15 days!
4. After the specified period, carry out the same steps, and the bootloader will get unlocked and you will see the result shown in Figure 3.

Installing Team Win Recovery Project (TWRP)

Team Win Recovery Project (TWRP) is an open source custom recovery image for Android based devices. It provides a touchscreen-enabled interface that allows users to install third-party firmware and back up the current system. It is installed on an Android device when flashing, installing or rooting Android devices.

1. Download the Pixel Experience ROM for your device from the